A sharper combined contraction frontier for exact SAT

Third round research report

27 October 2026

Abstract

The combined normalized polynomial-space certificate improves, but the logarithmic order does not. After reviewing the canonical project and eight new incoming archives, we combine the already supplied asymmetric balancing and integral slope-four Cauchy modes, then improve the resulting schedule. A depth-varying grouping alphabet gives

$$T(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)} 2^{(63.75735555\ldots+o(1))s/\sqrt{\lceil \log_2 s \rceil}}$$

with polynomial space, below the $66.12047546\ldots$ retuned two-stage comparison from the compatible supplied ingredients. Thus coefficient 64 is valid eventually on the shared wire-normalized interface. An optional subexponential-space cache gives eventual coefficients 38 for time and 34 for space. A stronger low-degree assumption gives polynomial-space coefficient 24, or cached time 14.3 and space 12.4. We also provide exact component improvements: sparse incremental conditional-expectation scores, antipodal centered packet extraction and a favorable-profile cube-root filter, with independent correctness checks and explicitly scoped local measurements. These are not a complete SAT implementation, a proof-assistant certificate, a numerical coefficient directly on the number of SAT variables, or a smaller unrestricted logarithmic-order bound. All SAT consequences retain the source manuscript's exact algebraic and reduction dependencies.

1 Results and scope

Let n be the number of declared variables, m the number of input clauses and $k = 3$ the fixed clause width. The source manuscript's general-density exponent is $O(n \log\log n/\sqrt{\log n})$, with $O_D(n/\sqrt{\log n})$ for $m \le Dn$ and fixed D [1, 3]. The supplied density refinements give the uniform envelope

$$\mathrm{poly}_k(n,m)\, 2^{O_k\left(\frac{n}{\sqrt{\log n}} \min\{\log(2+m/n), \log\log n\}\right)}. \tag{1}$$

The new results preserve this order. In particular, they do not establish $o(n \log\log n/\sqrt{\log n})$ for unrestricted 3-SAT.

The numerical certificates below use the same normalized contraction size s: a bipartite Boolean tensor network with at most s wires, polynomially many vertices, maximum degree at most $\left\lceil s^{9/10} \right\rceil$, and total description length $\mathcal{L}$. Parallel edges have separate ports. Entries lie in

$$(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad 1 \le P_0, \ell \le s.$$

Each original entry can be evaluated in $(s+\mathcal{L})^b 2^{As^{9/10}}$ time and $(s+\mathcal{L})^b$ space for fixed A, b. All arithmetic is exact. Fixed constant multiples in the precision bounds do not change the leading results. The reduction has $s = O_k(n \log\log n)$ in general; its hidden constants preclude copying the coefficient 64 onto $n \log\log n$.

What is new here. The main mathematical step is a sequence of different alphabets at successive depths, exploiting the evolving asymmetric degrees rather than paying for one root alphabet and one fixed tail alphabet. The practical steps refine two existing exact subroutines, not the contraction identities themselves. The sparse score update is compared with the already supplied row-sum-cached algorithm, not just the slower literal triple sum. The centered extraction identity is an application of classical polarization; its packet adaptation and exact 2-adic precision accounting are the contribution here.

What is inherited. Near-unit balanced packing, thin additive precision gaps, sharp Hermite taper loss, scalar parity lifts, density reductions, slope-four Cauchy modes, and cached block contraction were all in the supplied proposals. We verify compatibility rather than relabeling them as new. "Derived" throughout means an ordinary mathematical derivation relative to the unchanged source deformation, integrality, recovery and recursive-factorization identities. No upstream program or installer was run, no upstream repository was modified, and no complete SAT solver or proof-assistant build was executed.

2 Snapshot and proposal comparison

The arrival snapshot was

`d098ba436b33df19365a44dd178694ff52845d19`.

It was authored at 02:00:24 UTC on 27 October 2026. Compared with the prior final pin `047f4f0426fa`, it is 14 commits ahead with 277 changed files. Exactly eight changed files are in the ETH/SAT review scope, all ZIP additions under `docs/incoming`; the other changes concern the separate DiophantineConsistency workstream. The canonical `Computability/SubexponentialSAT/Research` tree is unchanged at `2de706601ded18cad212c77ffe0aff2b09d3d4c5`, as is its README. Its 150-entry recursive inventory was not truncated.

Seven ZIPs arrived in commit `3e57705cf14e` at 01:54:16 UTC, including our already delivered second-round report; the eighth is the matching-packing packet at the frozen HEAD. All eight downloaded archive bytes match the immutable directory listing's Git blobs. All 182 checksum-listed files in the seven genuinely new packets match; their seven manifests bring the extracted file count to 189. Imported programs were read only. This review covers the canonical project, every current incoming ETH/SAT archive and the complete change list, not a fresh full-content scan of unrelated repository files [4, 5].

2.1 Closing canonical placement

The closing source pin is

`474ddba696a5d385c091c23fb2995f69edd13e00`.

Authored at 02:11:13 UTC, it places all eight reviewed arrivals in the canonical Research subtree. Every one of the 211 newly filed files matches an authenticated archive member's Git blob. All earlier mathematical files are unchanged; only the administrative omission ledger and comparative README changed. The new Research tree has 405 entries and is not truncated. Forty-one archive members are omitted by the filing policy: three original checksum ledgers and 38 duplicate sources/documents. Only a README remains in `docs/incoming`; the eight archives remain available at their immutable arrival commits.

This relocation supplies no new theorem or implementation. The closing repository-wide comparison reached the API's 300-file cap, so it is not used to claim complete whole-repository coverage. The

complete target Research tree, project and incoming listings, and exact byte mapping establish the scoped conclusion. The final citations below use this closing canonical pin; archive acquisition records retain the arrival pin. This closes the observation window for the present deliverable.

2.2 The eight arrivals

Coefficients in this table multiply $s/\sqrt{\log_2 s}$ on the stated normalized interface. "Same order" means the general and fixed-density orders in Section 1 are unchanged. These are supplied mathematical certificates, not measured SAT runtimes.

Packet	Contribution already supplied	Displayed coefficient
Second-round report	Matching-potential packing, thin gaps, crossing accounting and root/tail scheduling; our previous delivered work.	84.0663, polynomial space.
Packet refinement	Two-stage balanced pair labels; $D_{\mathrm{arr}}\Delta \operatorname{polylog} s$ rather than $D_{\mathrm{arr}}^2\Delta$ padding condition; additive gap.	95.6639, polynomial space.
Recursive refinement	Sharper original packing; additive guard; actual-row basis with slope 64/21 rather than 3.	133.2074, polynomial space.
High moment	Even-moment conditional expectation; near-unit padding; separate orbit alternative.	94.7482; setup $2^{O(\log^2 s)}$. Orbit: 98.8967 with polynomial setup.
Joint Cauchy	Full orientation-half integral modes of slope four; unchanged family multiplicity and group locality. Optional cached three-block contraction.	319.6858 polynomial space; 225.5971 time and 196.1994 space with cache.
Balanced packing	Exact-rational two-stage moment-generating potential and adaptive integer capacities.	94.7482, polynomial space and setup.
Asymmetric balancing	Track old-side versus new-packet degrees; asymmetric pair-load bound; actual-row annulus bases.	76.9090 generic; 31.0641 only for $\Delta \le \left\lceil s^{1/12} \right\rceil$.
Matching and thin gaps	Exact potential over a proper edge coloring; near-perfect pair labels; integral thin-gap schedule.	94.7482, polynomial space and setup.

All seven genuinely new packets retain the same logarithmic order. Most compared only with the older canonical corpus, not with the other same-batch arrivals. Their quoted 648.4411 or 374.3776 baselines are therefore not the correct combined frontier for this iteration.

2.3 The unchanged earlier proposals

Write $L = \log n$ and $\ell = \log \log n$. These seven remain part of the comparison [9, 10].

Packet	Established refinement	Stronger target status
Factorial Cauchy	Weights $1 + 3b - 2s_2(b)$ and leading entropy factor $1/\sqrt{3}$.	Same order.

Packet	Established refinement	Stronger target status
Contraction	Exact Hermite count; leading precision $24MD/u \to 6MD/u$; streamed extraction. Coefficient 648.4411, or 374.3776 with factorial modes.	Same order; tuned family infimum about 368.6540.
SAT density	Uniform profile in equation (1), using local codes and actual query sparsity.	Improves intermediate density, not endpoint log powers.
Precision	Exact full-loss taper and minimax accounting.	$n\ell/L$ general and n/L sparse require structured-mode summation.
Scalar parity	Exact scalar lift, lower incidence degree, no truncated coefficient axis.	$n\ell^{3/2}/L$ needs compact factorization.
ETH density	Overlapping density-sensitive profile, retuned constants and local reciprocal Cauchy Smith invariants.	Not another independent density multiplier.
Prior asymptotic analysis	Scoped representation barriers and precise missing recursive interfaces.	$n\sqrt{\ell/L}$ remains conditional.

The original OpenAI PDF is the source manuscript, not another competing proposal. The separate 49-page manuscript attributed within the PDF to Claude/Anthropic states an exponent $O(\min\{k^2 n \log\log n/\sqrt{\log n}, km/\sqrt{\log m}\})$. It supplies no directly comparable explicit numerical constant, and its claimed Lean development is unavailable in the reviewed source. We therefore do not assert numerical domination of that algorithm or independent certification of its formalization. Organizational attribution printed inside a PDF is not independently authenticated authorship.

2.4 The combined baseline and resource distinctions

Near-unit padding, asymmetric degree tracking, thin precision gaps, sharp taper loss and slope-four modes act at different interfaces and can compose. The strongest displayed asymmetric coefficient 76.9089949516 becomes 66.6051434076 by the supplied slope-four substitution. Its own fixed-r excluded-boundary family gives a still stronger matched infimum 66.1927236936 at $r = 16$. Allowing different root and tail choices must also be counted when evaluating the combined envelope; Section 4 states the resulting comparison without claiming global schedule optimality.

The new 63.7574 certificate is not merely this algebraic substitution. It changes the depth schedule. The separate cached route in Section 5 pays subexponential memory and must be compared in that resource class. A polynomial-space result alone cannot claim to dominate a faster memory-intensive certificate on time.

The supplied low-degree 31.0641 coefficient assumes $\Delta \le s^{1/12}$, not merely fixed clause density. The density packet's parity-incidence reduction can have degree $s^{3/4+o(1)}$ and does not satisfy that premise. Direct bounded-degree CNF networks can satisfy it, but hidden reduction wire constants still prevent treating a wire coefficient as a coefficient on n.

3 An integral slope four oracle that composes with the schedule

The incoming joint-Cauchy packet already supplies the full all-level construction [6]. The older density packet already contains the local odd-Cauchy Smith valuations. Neither result is new here. The purpose of this section is to check that the stronger basis remains integral on the entire interpolation domain and preserves the one-source/one-target recursive interface used by the new schedule.

3.1 Telescoping on the actual orientation half

At a retained interpolation node, let $L = 2^a$, $i = v_2(t)$, $R = 2^i$, $N = L/2$, $T = t + 2^{a+1}\zeta$, and $\Gamma = C(T)/C'(0)$ in the source's unramified extension ring $\mathcal{O}$ of $\mathbb{Z}_{(2)}$. Here C is the source interpolation polynomial and $\zeta \in \mathcal{O}$. Put

$$s_j = (j \bmod R) + 2R\lfloor j/R \rfloor, \qquad 0 \le j < N.$$

These are exactly the slots with bit i zero; adding R gives the other orientation. This offset cancels in the crossing difference. Every orientation half belongs to one source group and its crossing rows to one target group. Define

$$P_b(x) = \prod_{h<b} \frac{x - s_h}{T + x - s_h}, \qquad Q_b(y) = \prod_{h<b} \frac{s_h - y}{T + s_h - y},$$
$$f_b(x) = \frac{TP_b(x)}{T + x - s_b}, \qquad g_b(y) = \frac{TQ_b(y)}{T + s_b - y}.$$

The cross-multiplied identity

$$(T + x - s_b)(T + s_b - y) - (x - s_b)(s_b - y) = T(T + x - y)$$

telescopes the remainders $P_b(x)Q_b(y)/(T + x - y)$, giving

$$\frac{1}{T + x - y} = \frac{1}{T} \sum_{b=0}^{N-1} f_b(x) g_b(y). \tag{2}$$

The final remainder vanishes because x is a pivot slot. This is a finite identity, not a convergence argument or an appeal to an unevaluated Smith normal form.

Write $b = qR + r$ with $0 \le r < R$, and put $E_b = q + v_2(q!)$. For $x = s_j$, $j \ge b$, a predecessor with different residue modulo R has equal numerator and denominator valuations. The predecessors with the same residue are $\ell R + v$, $0 \le \ell < c$, with $c = q$ or $q + 1$. Their numerator product has valuation

$$c(i + 1) + v_2(k(k - 1) \cdots (k - c + 1)),$$

whereas every denominator has valuation i. Since $k \ge c$ and $\binom{k}{c}$ is an ordinary integer, the net contribution is at least $c + v_2(c!) \ge E_b$. The final T/denominator factor is integral. The same argument applies to g_b; both vanish for $j < b$. At $j = b$, the valuation is exactly E_b.

Consequently $F_b = 2^{-E_b} f_b$ and $G_b = 2^{-E_b} g_b$ are integral, triangularly supported features with unit diagonal. Setting

$$A_b(x) = \frac{\Gamma}{T} 2^{2E_b} F_b(x)$$

gives the exact crossing decomposition $W_{xy} = \sum_b A_b(x) G_b(y)$ and

$$v_2(A_b(x)) \ge a + 1 - i + 2q + 2v_2(q!) = a + 1 - i + 4q - 2s_2(q). \tag{3}$$

For every q, there are exactly R remainders r per orientation. Two orientations yield $2R$ modes per q, the same multiplicity as the old tile families. A source form now reads a whole half-packet rather than one tile, but still only one group. Missing real slots are simply omitted; pivots need not contain real ports.

The divisibility argument uses ordinary integer slot differences, not an assertion that binomial polynomials of arbitrary extension elements are integral. It therefore holds for every $\zeta \in \mathcal{O}$, including zero and nonunits.

3.2 Residue evaluation without nonunit division

For a slot difference d, let $\nu = i$ if $R \mid d$, and $\nu = v_2(d) < i$ otherwise. Compute the normalized denominator directly as

$$(T + d)/2^{\nu} = (t + d)/2^{\nu} + 2^{a+1-\nu}\zeta,$$

which is a unit. Split every nonzero integer numerator into its exact power of two and signed odd part before reduction. Multiply odd parts and unit inverses and add the exponents; the integrality proof guarantees that subtracting E_b leaves a nonnegative exponent. This computes F_b and G_b modulo 2^p without dividing a reduced residue by a power of two. Return zero first for $j < b$.

There is also a same-precision prefactor formula. For $v \in [-L, L-1]$, $v \neq t$, write $n_v = v_2(t - v)$. Grid completeness gives $\sum_{v \neq t} n_v = v_2(C'(0))$. With $c_{\mathrm{odd}} = C'(0)/2^{v_2(C'(0))}$,

$$\begin{aligned}\frac{\Gamma}{T} &= 2^{a+1-i}\zeta\,(T/2^i)^{-1}c_{\mathrm{odd}}^{-1} \\ &\quad \cdot \prod_{v \neq t}\left[(t - v)/2^{n_v} + 2^{a+1-n_v}\zeta\right].\end{aligned}$$

Every inverse is of a unit. The formula handles nonunit ζ without extracting its exact valuation. No factorial inverse, ambiguous lift, or hidden temporary precision loss is used.

Each feature takes $O(L)$ ring operations. Since L and all source precision and extension parameters remain polynomial in s, the local descriptions and feature evaluation remain polynomial-size. The outer packet-entry cost remains the inherited $\exp(O(L \log s))$ bound, or a correctly guarded extraction alternative. Immutable creation-time representatives are unchanged; descriptions are not recursively composed.

3.3 Uniform entropy and recursive compatibility

Each mode still has one source group and one target group, so the child-network identity and port degrees remain intact. A coefficient indexed by multiplicities β has valuation at least $\sum_b w_b\beta_b$; tuples of weight at least p vanish exactly modulo 2^p.

For padded mass $B = (1 + o(1))M$, alphabet r, and $D = 2^{a-i}$, unequal-symbol packets contribute at most

$$c \le \big(2(1 - 1/r) + o(1)\big)M/D$$

mode families. Lowering the actual weights to $1 + 4q - 2s_2(q)$ only increases the count. Their generating product has logarithm $(\pi^2/(24\theta) + o(1/\theta))$ per family. Combined with the inherited precision envelope $P \le (6 + o(1))MD/u$, optimization of the positive parameter θ yields

$$\log_2 N \le (K_4(r) + o(1))M/\sqrt{u}, \qquad K_4(r) = \frac{\pi}{\ln 2}\sqrt{2(1 - 1/r)}. \tag{4}$$

These are upper envelopes, not claims that every instance attains the maximal family count or precision. The thin gap ensures D dominates every polylogarithm needed for uniform small-saddle error. Relative to slope three, the leading coefficient is multiplied by $\sqrt{3}/2$; this does not change the logarithmic order.

An independent standard-library checker evaluates the actual ring $(\mathbb{Z}/2^{37})[U]/(U^2+U+1)$ over all retained levels for $1 \le a \le 5$, positive and negative nodes, and five values $\zeta = 0, 1, U, 1+U, 2+3U$. Normal and optimized runs agree on 200 configurations, 22,190 matrix identities, 110,950 strict-budget truncations, 44,380 integral features, 3,560 unit diagonals and 200 missing-slot bilinear factorizations. Each attempted inverse rejects a nonunit explicitly. The proof above establishes the general result; the tests check its local implementation.

4 A variable alphabet contraction schedule

The following schedules improve the *combined* frontier, not just any single supplied report. We reuse the supplied asymmetric pair-load lemma, thin precision gaps, exact Hermite taper, and all-level full-block Cauchy factorization. The new ingredients here are a finite succession of grouping alphabets, a carefully allocated geometric tail, and, for low-degree inputs, allowing generated packet degrees to grow while charging their actual size. No optimality claim is made for the schedules.

4.1 The local bound and what is being compared

Put $\lambda = \lceil \log_2 s \rceil$. At depth d, let u_d be the retained valuation range, a_d the packet exponent, M_d the active edge count, and r_d the grouping alphabet. The compatible supplied ingredients give

$$M_{d+1} \le (2/r_d)(1+o(1))M_d, \tag{5}$$

$$\log_2 N_d \le (K(r_d)+o(1))M_d/\sqrt{u_d}, \qquad K(r) = \frac{\pi}{\ln 2}\sqrt{2(1-1/r)}. \tag{6}$$

All error terms below are uniform over active calls. To see the coefficient, write $D = 2^{a_d - i}$. The precision envelope is $(6+o(1))M_d D/u_d$; the full-block mode-family envelope is $(2(1-1/r_d)+o(1))M_d/D$. The joint weights have slope four: $w_j = a_d + 1 - i + 4j - 2s_2(j)$. Their partition product is bounded above by that for $4j + 1 - 2s_2(j)$, whose logarithm is $(\pi^2 + o(1))/(24\theta)$ per family. Optimizing the positive real parameter θ yields (6). Dropping the positive intercept is an upper bound; no assertion about a sharp joint limit of the intercept and saddle is needed.

For comparison, the supplied asymmetric schedule has leading coefficient $76.9089949516\ldots$ with slope-three modes. Simply substituting the supplied joint modes gives $66.6051434076\ldots$, without any new schedule. The asymmetric report also gives an excluded-boundary constant $76.4327736858\ldots$; multiplying this stronger comparison by $\sqrt{3/4}$ gives $66.1927236936\ldots$. More explicitly, its limiting $r = 16$ root budget is $1/80$ and the remaining budget is $1/80$; optimally allocating the latter geometrically gives

$$\frac{K(16)}{\sqrt{1/80}} + \frac{(1/8)K(16)}{\sqrt{1/80}(1-(1/8)^{2/3})^{3/2}} = 66.1927236936\ldots.$$

This boundary is an infimum of admissible fixed-slack schedules, not an attained finite parameter choice. The new bound below is smaller even than it. A stronger retuned comparison allows $r_0 = 17$ at the root and $r = 13$ thereafter. With $V = 1/(20\log_2 17)$, $B = 1/(10\log_2 13) - V$, its excluded-boundary value is

$$\frac{K(17)}{\sqrt{V}} + \frac{(2/17)K(13)}{\sqrt{B}(1-(2/13)^{2/3})^{3/2}} = 66.1204754600\ldots.$$

The new theorem also improves on this more generous comparison. We do not claim this two-stage value is globally minimal or that all possible recombinations of the supplied ideas have been optimized. These comparisons concern upper-bound certificates, not lower bounds on previous algorithms.

Theorem 4.1 (Generic polynomial-space coefficient below 64)**.** *On the original normalized interface with at most s edges, maximum degree at most $\lceil s^{9/10} \rceil$, $P_0, \ell \le s$, polynomial description length dependence, and original entry allowance* $\mathrm{poly}(s+\mathcal{L})2^{As^{9/10}}$ *for fixed A, the unchanged source identities together with the supplied local lemmas admit contraction in polynomial space and time*

$$(s+\mathcal{L})^{O(1)} 2^{(63.7573555464258\ldots+o(1))s/\sqrt{\lambda}}.$$

In particular the coefficient can be replaced by 64 *for sufficiently large s. This is a relative contraction theorem, not an independent verification of the whole source or a coefficient directly on SAT variables.*

4.2 An entirely rational schedule

Use $r_0, \ldots, r_5 = (13, 14, 8, 6, 5, 4)$ and $r_d = 3$ for $d \ge 6$. Define the rational cumulative leading budgets

$$(A_0, \ldots, A_5) = (.01351, .02626, .03333, .03868, .04306, .04999), \qquad A_* = .06308, \qquad q = \frac{763}{1000}.$$

Every displayed terminating decimal here denotes its exact rational value. Put $A_{-1} = 0$, $v_d = A_d - A_{d-1}$ for $0 \le d \le 5$, and

$$v_{6+j} = (A_* - A_5)(1-q)q^j \quad (j \ge 0), \qquad u_d = \lfloor \lambda v_d \rfloor.$$

Set

$$g = \lceil \log_2 \lambda \rceil^2, \qquad a_d = \sum_{j=0}^{d} u_j + (d+1)g, \qquad H = \lceil s/\lambda \rceil.$$

Below a finite threshold, or if the required integer inequalities fail, use direct contraction. Thus no small input is forced into a degenerate zero-range taper. The fixed rational slack is deliberate: the theorem does not rely on implementing an excluded boundary or using floating-point logs.

Proof of Theorem 4.1. We first establish uniform applicability without assuming a recursion depth. Consider every hypothetical depth $0 \le d \le D_* := \lceil \log_{10/7} \lambda \rceil + 10$ and every active $M > H$. In this range $(d+1)g = O((\log \lambda)^3) = o(\lambda)$. For the six finite stages, $a_d \le A_d \lambda + o(\lambda)$; in the tail, $a_d \le A_* \lambda + o(\lambda)$. Every u_d tends to infinity uniformly: indeed $q > 7/10$ gives

$$\min_{d \le D_*} u_d \ge \lambda^{\eta + o(1)}, \qquad \eta = 1 + \frac{\ln q}{\ln(10/7)} > 0.$$

Floors are consequently a uniform relative $o(1)$ perturbation.

At the root, symmetric packing needs $r_0^{2a_0} s^{9/10}\,\mathrm{poly}(\lambda) = o(M)$. The leading inequality is $20A_0 \log_2(13) < 1$. Below the root, the supplied child-side invariant is $R_d \le \lceil s^{9/10} \rceil$ and $C_d \le 2^{a_{d-1}+1}$. The asymmetric packing lemma requires

$$Q_d R_d / M = o(1), \quad Q_d^2 C_d \lambda^2 / M = o(1), \quad 2^{a_d} Q_d^2 / M = o(1), \qquad Q_d = r_d^{a_d}.$$

For the row term it suffices that $10A_d \log_2 r_d < 1$ at finite stages $d \ge 1$, and $10A_* \log_2 3 < 1$ in the tail. Every one of these inequalities is strict and checked by integer powers: if $A = n/m$, check $13^{20n} < 2^m$ at the root and $r^{10n} < 2^m$ thereafter. No approximate logarithm is used in these checks. The other tail exponents are smaller than $A_* + 1/5 = .26308 < 1$. The root rounding exponent is less

than $A_0 + 1/10 < 1$. All logarithmic factors, $s/M \le \lambda$, ceilings, and the additive precision gaps are absorbed by the fixed positive margins. Also $2^{a_d+1} < s^{9/10}$ eventually, so no created tensor violates the high-degree premise. Uniform near-unit padding follows.

For sufficiently large s, each hypothetical child shrinks by at least a factor $7/10$, since the largest limiting ratio is $2/3$. An active recursion path cannot reach D_*. This closes the induction and validates the hypothetical-depth argument. More sharply, the padding bound is $1 + O(1/\lambda) + s^{-\epsilon+o(1)}$ for a fixed $\epsilon > 0$; its product over $O(\log \lambda)$ depths is $1 + o(1)$. Therefore, with $m_0 = 1$ and $m_d = \prod_{j<d}(2/r_j)$,

$$M_d \le (1 + o(1))m_d s.$$

The inherited precision is at most $s\lambda^{O(1)}2^{a_{d-1}}$. Since every retained $i < u_d$ has $2^{a_d-i} \ge 2^{a_{d-1}+g+1}$, its ratio to $M_d 2^{a_d-i}/u_d$ is at most $\lambda^{O(1)}2^{-g} = o(1)$. The evaluation loss is negligible by the same estimate. Thus the supplied exact taper yields the uniform six-bit envelope used in (6). All old tensor descriptions retain their creation precisions and canonical representatives; the schedule changes none of these algebraic conventions.

Summing (6) along a worst path gives the coefficient

$$C = \sum_{d=0}^{5} \frac{m_d K(r_d)}{\sqrt{v_d}} + \frac{m_6 K(3)}{\sqrt{(A_* - A_5)(1-q)}(1 - (2/3)/\sqrt{q})} = 63.7573555464258\ldots. \tag{7}$$

The denominator is positive because $q > (2/3)^2$. The six finite contributions are, respectively, 52.98228468, 8.41547470, 1.56719246, 0.43954213, 0.15865534, and 0.04885070; the infinite tail contributes 0.14535554. These decimals explain the allocation but are not proof inputs. The accompanying exact-rational checker bounds square roots by integer squares, π by alternating Machin-series bounds, and $\ln 2$ by its positive $2\,\mathrm{atanh}(1/3)$ series with an explicit tail bound. It certifies $C < 64$.

There are polynomially many interpolation nodes, values, and child-group choices per call. Across $O(\log \lambda)$ depths their logarithmic overhead is $O(\lambda \log \lambda) = o(s/\sqrt{\lambda})$. Direct leaves cost $2^{O(s/\lambda)}$. Entry work is bounded by the original $2^{O(s^{9/10})}$ allowance and packet work $\exp(O(2^{a_d} \log s)) = \exp(s^{A_*+o(1)})$; both are lower order. All precision, arithmetic, label-set, and deformation-degree parameters remain polynomial in s; in particular $M(2L-1) < s^4$ eventually. Depth-first accumulation with immutable descriptions gives polynomial space. The dependence on description length stays a fixed polynomial because branching depends on s, not that length. This proves the theorem. □

4.3 Tracking generated degrees improves the low-degree coefficient

The supplied low-degree theorem holds all generated degrees below $s^{1/12}$. This is sufficient but unnecessary: a later call can pay for the actual degree of its immediately preceding packet scale instead. The fact that all retained vertices move to one side does not create new incident ports. Explicitly, the fresh side has $C_d \le 2L_{d-1}$, while the retained side has $R_d \le \max\{\lceil s^{1/12} \rceil, 2L_{d-2}, 2L_{d-3}, \ldots\}$; in particular both sides are bounded by $s^{\max\{1/12, A_{d-1}\}+o(1)}$. Consequently, when packet exponents are nondecreasing and exceed the initial degree exponent, *both* child sides have maximum degree at most $2^{a_{d-1}+1}$: retained older packets are no larger than the last ones, and original vertices are smaller still. Symmetric matching packing already suffices for the following theorem; asymmetric packing is not needed in this smaller-degree branch.

Theorem 4.2 (Low-degree polynomial-space coefficient below 24)**.** *Under the same normalized interface, but with root maximum degree at most $\lceil s^{1/12} \rceil$, there is polynomial-space contraction in time*

$$(s + \mathcal{L})^{O(1)} 2^{(23.6661688071455\ldots+o(1))s/\sqrt{\lambda}}.$$

Thus the coefficient 24 *is eventually valid. This premise holds for the direct bounded-degree sparse-CNF network, not for arbitrary dense incidence networks merely because they arise from SAT.*

Proof. Use $r_0, \dots, r_5 = (35, 11, 7, 6, 5, 4)$, then $r_d = 3$, with

$$(A_0, \dots, A_5) = (.08881, .12627, .15116, .16206, .17717, .19999), \qquad A_* = .2398.$$

Use exactly the preceding rational q, definitions of v_d, u_d, a_d, and leaf threshold. Here $A_0 > 1/12$ and all A_d increase strictly, so after the root the maximum degree is at most $2^{a_{d-1}+1}$ as just explained. Every finite stage and the tail satisfy

$$A(1 + 2\log_2 r) < 1.$$

For $A = n/m$, the exact integer check is $r^{2n} < 2^{m-n}$. These inequalities control both rounding LQ^2 and the packing term $Q^2\Delta$: below the root its leading exponent is $2A_d \log_2 r_d + A_{d-1} < A_d(1+2\log_2 r_d)$, and at the root $2A_0 \log_2 r_0 + 1/12 < A_0(1 + 2\log_2 r_0)$. All inequalities have fixed strict margins. The same noncircular depth, uniform floor, precision, and path estimates apply. The maximum packet exponent is now $.2398\lambda + o(\lambda)$, still smaller than $.9\lambda$ and well within polynomial arithmetic and lower-order entry work.

Formula (7) with these parameters is $23.6661688071455\ldots < 24$, certified by the same exact arithmetic. The six finite contributions are 21.19892854, 1.80435144, 0.39079845, 0.16636684, 0.04614898, and 0.01454394; the geometric tail adds 0.04503062. For comparison, combining joint modes with the supplied low-degree $31.0641320630\ldots$ bound gives $26.9023275131\ldots$. Again the new theorem improves the combined bound, not just its slope-three component. The source SAT reduction constants are still needed to convert any of these wire-level coefficients to a coefficient on n. □

4.4 What these schedules do not establish

The generic and low-degree certificates preserve the $s/\sqrt{\log s}$ scale. General-density reductions still carry their $O(n \log\log n)$ wire budget; these proofs neither remove that factor nor provide a smaller unrestricted logarithmic order. Neither coefficient is a measured running time or a formalized end-to-end SAT theorem. The auxiliary search explored a restricted saturated-milestone schedule family; its numerical candidates are not optimality certificates. The rational schedules and all-size arguments above stand independently of that search.

5 A separate time and space tradeoff

The incoming joint-Cauchy packet already supplies exact-weight mode allocations, cached group functions, three-block boundary tables and division-free Strassen multiplication. This section changes the alphabet, taper and block sizes; it does not claim a new matrix-multiplication identity or treat the cache as polynomial space. The generic normalized input interface is the same as in Section 1.

Theorem 5.1 (A cached certificate below 38 time and 34 space)**.** *Relative to the unchanged source identities and supplied integral crossing factorization, the following schedule yields*

$$T(s, \mathcal{L}) \le (s + \mathcal{L})^{O(1)} 2^{(37.9301090828846\ldots + o(1))s/\sqrt{\lambda}},$$
$$S(s, \mathcal{L}) \le (s + \mathcal{L})^{O(1)} 2^{(33.6230409216578\ldots + o(1))s/\sqrt{\lambda}}, \qquad \lambda = \lceil \log_2 s \rceil .$$

Thus eventual coefficients 38 *and* 34 *are valid. This is a separate subexponential-space tradeoff, not the polynomial-space theorem of Section 4.*

5.1 Exact weight allocation and block costs

Equal pair-label padding bounds the number of crossing mode families on each ordered unequal group pair by $(2+o(1))M/(r^2D)$. For each exact per-pair weight allocation below the precision budget, let h_e be the binary length of its pair-mode list. The supplied weighted generating-function argument gives, for fixed nonnegative pair weights w_e,

$$\sum_e w_e h_e \le \left(\sqrt{\sum_e w_e^2/r^2} + o(1)\right) \mathcal{K}(M,u), \qquad \mathcal{K}(M,u) = \frac{\pi\sqrt{2}}{\ln 2}\frac{M}{\sqrt{u}}. \tag{8}$$

The number of weight allocations is polynomial for fixed r and polynomial precision; empty lists are discarded. A common global weight budget is essential: independently assigning each pair the full precision would overcount. Group functions are recursively computed and cached, not assumed to have inexpensive entries.

Partition the r groups into three blocks of sizes k_1, k_2, k_3. A group is incident with $2(r-1)$ ordered pairs, so the coefficient for computing its cache is

$$\alpha_r = \frac{\sqrt{2(r-1)}}{r}.$$

A block of size k touches $2rk - k^2 - k$ pair lists, while its boundary has $2k(r-k)$ lists. Hence the maximal table-building and storage factors are

$$A_r = \max_j \frac{\sqrt{2rk_j - k_j^2 - k_j}}{r}, \qquad J_r = \max_j \frac{\sqrt{2k_j(r-k_j)}}{r}.$$

Set $v_{ij} = 2k_ik_j$ and $\tau = \log_2 7 - 2$. Rectangular division-free Strassen multiplication has factor

$$B_r = \frac{1}{r}\sqrt{v_{12}+v_{13}+v_{23} - (1-\tau^2)\min(v_{12},v_{13},v_{23})}, \qquad \beta_r = \max(A_r, B_r).$$

These follow by applying equation (8) to the relevant pair-list weights. The seven-product formula uses no division and therefore works over the source quotient ring. Sequential record access, sorting and reblocking follow the supplied implementation accounting. Suppressing polynomial factors, the time recurrence is

$$T_d \le 2^{(\alpha_{r_d}+o(1))\mathcal{K}_d} r_d T_{d+1} + 2^{(\beta_{r_d}+o(1))\mathcal{K}_d}. \tag{9}$$

The second term reads already computed group caches; it does not pay a new child contraction per table entry.

For $r = 7$ use blocks $(3,2,2)$, giving

$$\alpha_7 = \sqrt{12}/7, \quad \beta_7 = \sqrt{30}/7, \quad J_7 = \sqrt{24}/7.$$

The matrix term $\sqrt{24+8\tau^2}/7$ is smaller than the table-building term. For $r = 9$ use $(3,3,3)$:

$$\alpha_9 = 4/9, \quad \beta_9 = \sqrt{36+18\tau^2}/9, \quad J_9 = 2/3.$$

Here the matrix term exceeds $\sqrt{42}/9$.

5.2 A rational root and tail schedule

Choose

$$r_0 = 7, \quad u_0 = \lfloor 89\lambda/5000 \rfloor; \qquad r_d = 9, \quad u_d = \left\lfloor \frac{1233}{125000}(7/25)^{d-1}\lambda \right\rfloor \quad (d \geq 1).$$

The cumulative leading budget is

$$\frac{89}{5000} + \frac{1233/125000}{1 - 7/25} = \frac{63}{2000}.$$

Use the same thin guard $g = \lceil \log_2 \lambda \rceil^2$ and cumulative $a_d = \sum_{j \leq d} u_j + (d+1)g$, with leaf threshold $H = \lceil s/\lambda \rceil$. The larger polylogarithmic guard is chosen conservatively to make every inherited polynomial factor negligible.

At the root, symmetric packing is valid because $.9 + 2(89/5000)\log_2 7 < 1$, witnessed exactly by $7^{89} < 2^{250}$. Below the root, the asymmetric side invariant is $R_d \leq \lceil s^{.9} \rceil$, $C_d \leq 2^{a_{d-1}+1}$; the row term is controlled by $.9 + (63/2000)\log_2 9 < 1$, witnessed by $9^{63} < 2^{200}$. The other packing and rounding terms have leading exponent at most

$$(63/2000)(1 + 2\log_2 9) < .232 < 1.$$

New degrees remain below $s^{.9}$.

For a noncircular bootstrap, consider every candidate depth $d \leq \lceil \log_4 \lambda \rceil + 3$. The guards sum to $O((\log \lambda)^3) = o(\lambda)$, all strict margins hold uniformly, and tail children eventually have at most $M/4$ edges. Thus no active path exceeds that range. Since $7/25 > 1/4$, every active u_d grows as a positive power of λ, so floors are negligible. More sharply,

$$M_d \leq (1 + o(1))s(2/7)(2/9)^{d-1} \quad (d \geq 1).$$

The thin guard gives the same uniform precision envelope $(6 + o(1))MD/u$. Labels, precisions and descriptions remain polynomial-size; original-entry work, packet-entry work, leaves and branching remain lower-order on the target scale. Creation-time representatives are unchanged.

5.3 Solving the recurrence and checking the constants

Let $\kappa = \pi\sqrt{2}/\ln 2$ and define the local coefficients

$$K_0 = \frac{\kappa}{\sqrt{89/5000}}, \qquad K_1 = \frac{(2/7)\kappa}{\sqrt{1233/125000}}, \qquad \gamma = \frac{2/9}{\sqrt{7/25}} = \frac{10}{9\sqrt{7}} < 1.$$

The tail local coefficient at depth $d \geq 1$ is $K_1\gamma^{d-1}$. Work stopping after j tail levels has coefficient

$$\alpha_9 K_1 \frac{1 - \gamma^j}{1 - \gamma} + \beta_9 K_1 \gamma^j.$$

It interpolates between two endpoints, so the tail coefficient is $K_1 \max\{\beta_9, \alpha_9/(1-\gamma)\}$. The first is larger: $\alpha_9/(1-\gamma) = .7662314\ldots < \beta_9 = .7676547\ldots$. Including the root gives

$$C_T = \max\{\beta_7 K_0, \alpha_7 K_0 + \beta_9 K_1\} = 37.9301090828846\ldots.$$

The root-only term is $37.5917025555\ldots$. Ancestor tables add rather than multiply memory, and the tail storage decreases geometrically. Therefore

$$C_S = \max\{J_7 K_0, J_9 K_1\} = 33.6230409216578\ldots.$$

One may retain the unreduced maximum for the time formula; the rounded bound does not need the displayed decimal comparison. An authored exact-rational checker uses $\pi < 22/7$, a positive-series lower bound for $\ln 2$, the integer witness $7^{100} < 2^{281}$ for $\log_2 7 < 2.81$, and square-root enclosures. It certifies $C_T < 37.956800 < 38$ and $C_S < 33.636580 < 34$. Floating point only displays the sharper formulas. Search suggestions near 37.8906 are not optimality certificates and are not substituted for this feasible rational schedule.

5.4 A cached low degree specialization

If the root maximum degree is at most $\left\lceil s^{1/12} \right\rceil$, retained tensors never gain ports. Both child sides therefore have degree at most $\max\{\left\lceil s^{1/12} \right\rceil, 2^{a_{d-1}+1}\}$. The invariant need not keep all later degrees below $s^{1/12}$. For increasing a_d with leading root exponent above $1/12$, symmetric matching packing is valid whenever $a_d(1 + 2\log_2 r_d) < \lambda$ has fixed leading slack.

Use $r_0 = 12$, $u_0 = \lfloor 239\lambda/2000 \rfloor$, followed by

$$r_d = 7, \qquad u_d = \left\lfloor \frac{567}{32000}(7/16)^{d-1}\lambda \right\rfloor \quad (d \geq 1).$$

The total leading budget is $151/1000$. The exact witnesses

$$12^{478} < 2^{1761}, \qquad 7^{302} < 2^{849}$$

certify the root and tail rounding/packing capacities. Created degrees are at most $s^{\cdot 151+o(1)} < s^{\cdot 9}$. Tail children have limiting ratio $2/7$ and eventual coarse ratio $1/3$. Candidate depth $\lceil \log_3 \lambda \rceil + 3$ and $7/16 > 1/3$ give the same noncircular bootstrap, growing tapers and negligible floors; the preceding precision and bit-cost argument applies unchanged.

Use blocks $(4, 4, 4)$ at the root. Then

$$\alpha_{12} = \sqrt{22}/12, \qquad \beta_{12} = \sqrt{2(2+\tau^2)}/3, \qquad J_{12} = 2/3.$$

The tail uses the $r = 7$ factors above. With

$$K_0 = \frac{\kappa}{\sqrt{239/2000}}, \qquad K_1 = \frac{\kappa/6}{\sqrt{567/32000}}, \qquad \gamma = \frac{8}{7\sqrt{7}},$$

the exact recurrence gives

$$C_T^{\text{low}} = \max\left\{\beta_{12}K_0,\ \alpha_{12}K_0 + K_1 \max\left(\beta_7, \frac{\alpha_7}{1-\gamma}\right)\right\} = 14.2391881944\ldots,$$
$$C_S^{\text{low}} = \max\{(2/3)K_0, J_7K_1\} = 12.3613014576\ldots.$$

Thus eventual coefficients 14.3 time and 12.4 space are valid on this additional low-degree interface. The exact checker proves the stronger rational upper bounds 14.251030 and 12.366279, respectively. This is not a generic-density result and remains a memory-intensive alternative.

For transparency, the finite block counts used throughout this section are:

Alphabet and blocks	Group incidence	Max block incidence	Pair classes	Max boundary
7 with $(3, 2, 2)$	12	30	$12, 12, 8$	24
9 with $(3, 3, 3)$	16	42	$18, 18, 18$	36
12 with $(4, 4, 4)$	22	76	$32, 32, 32$	64

Every squared factor is divided by r^2, not $r(r-1)$. Absent diagonal families reduce the number of summands; they do not redistribute their capacity onto unequal pairs.

5.5 Comparison with the full fixed alphabet cached family

A fair comparator is not merely the joint packet's old 225.5971 schedule, nor the asymmetric $r = 16$ example after adding a cache. Retuning all the supplied ingredients with fixed alphabet $r = 8$, blocks $(2, 3, 3)$, root taper $333\lambda/20000$, and tail tapers

$$(999/100000)(2/5)^{d-1}\lambda \quad (d \geq 1)$$

gives a feasible combined time coefficient $38.1868438947\ldots$ and space coefficient $34.0097101721\ldots$. Its total budget is $.0333\lambda$; the root, retained-side and rounding capacities have strict slack. The same guards and floors are admissible. Independent rational bounds certify eventual time 38.2 and space 34.1.

There is also a useful rigorous separation from an entire stated certificate family. Fix the supplied asymmetric/slope-four engine and its direct three-block construction plus seven-product cost model. Allow every fixed integer alphabet $r \geq 3$, every positive three-block partition and every positive taper allocation in this model. For generic degree exponent .9, its leading root and cumulative retained-side budgets obey

$$A_0 \leq \frac{.05}{\log_2 r}, \qquad A \leq \frac{.1}{\log_2 r}.$$

Put $c = 2/r$. Every time-certificate expression is at least its root-table term $\beta_r\kappa/\sqrt{A_0}$ and its complete cache-chain term. Relaxing the root cap in the latter and applying Hölder gives

$$\sum_{d\geq 0} \frac{c^d}{\sqrt{v_d}} \geq \frac{1}{\sqrt{A(1-c^{2/3})^3}}, \qquad \sum_d v_d \leq A,$$

so another lower bound on the certificate expression is $\alpha_r\kappa/\sqrt{A(1-c^{2/3})^3}$.

An independent exact-rational audit proves that every member of this family has coefficient strictly greater than 38:

- For $r = 3, \ldots, 7$, the relaxed cache-chain expression alone exceeds 38.
- For $r = 8, \ldots, 11$, enumeration of every unordered positive three-block partition shows the root-table expression exceeds 38.
- For $r \geq 12$, some block has $k \geq r/3$. Since $2rk - k^2 - k$ is increasing for integer $k < r$, its build factor satisfies $A_r^2 \geq 5/9 - 1/(3r) \geq 19/36$, while $\log_2 r > 7/2$. Conservative rational transcendental bounds make the root expression exceed 38 for every remaining integer r.

The weakest certified lower bound is $38.0299943343\ldots$, whereas the new rational schedule has certified upper bound $37.9567995131\ldots$. This establishes a strict separation without depending on a floating-point optimizer. The lower bound concerns these worst-case *upper-certificate expressions*, not actual algorithm runtimes, tensor contraction or SAT. It does not exclude other mode evaluators, matrix algorithms, cancellation methods or multistage schedules. The optional low-degree comparison is likewise scoped: applying the existing cache to the supplied low-degree $r = 48$ schedule gives 18.4107 time and 15.9887 space, but no global low-degree optimum is claimed.

6 Sparse incremental conditional expectation

6.1 The baseline already caches row sums

The incoming packet-refinement code already stores row power sums and suffix products; it compares Q candidate labels for each of n_c column vertices in $O(n_c Q^2)$ rational operations. Comparing only with

the asymmetric packet's literal triple sum would overstate the novelty. Our additional step removes the candidate-independent term and visits only row labels incident to the current column. For matched z, row labels and processing order, it preserves the exact minimizers, including first-label tie breaking, and therefore inherits the same pair-load theorem. It does not improve the padding bound.

Fix the row labels, a positive rational $z > 1$, and a column order. Let w_{aj} be the edge multiplicity between row label a and column vertex j. Let X_{ab} be the load assigned so far to the label cell (a, b). Define

$$S_a = \sum_{b=0}^{Q-1} z^{X_{ab}}, \quad f_{aj} = \frac{Q - 1 + z^{w_{aj}}}{Q}, \quad U_a = \prod_{k \geq j} f_{ak}, \quad E = \sum_a U_a S_a.$$

Here E is the conditional expectation of $\sum_{a,b} z^{X_{ab}}$, because the remaining columns are independently uniform. Parallel edges are bundled into the multiplicity; they are not independent labeling choices.

Proposition 6.1 (Sparse exact candidate scores)**.** *Put $A_j = \{a : w_{aj} > 0\}$ and $U'_a = U_a / f_{aj}$ on A_j, leaving $U'_a = U_a$ outside it. Define*

$$B = E + \sum_{a \in A_j} (U'_a - U_a) S_a,$$
$$\delta_b = \sum_{a \in A_j} U'_a z^{X_{ab}} (z^{w_{aj}} - 1).$$

Then assigning column j label b gives conditional expectation exactly $B + \delta_b$.

Proof. Remove the remaining random factor f_{aj} from each row. Assigning label b multiplies its one cell by $z^{w_{aj}}$ and leaves all other cells unchanged. The new row sum is $S_a + z^{X_{ab}}(z^{w_{aj}} - 1)$. Summing with U'_a yields the displayed identity. Every f_{aj} is positive, so all rational divisions are defined. Averaging δ_b over b and using $Q f_{aj} = Q - 1 + z^{w_{aj}}$ recovers the old E. Thus the least b minimizing δ_b preserves the usual conditional-expectation nonincrease invariant and exactly matches the dense method. Update $E \leftarrow B + \delta_b$, $U \leftarrow U'$, and only the affected S_a, X_{ab}. Empty columns choose label zero and leave E unchanged. Induction proves every maintained identity. □

6.2 Operation counts and bit costs

Let $D = \sum_j |A_j|$. With powers $z^0, \ldots, z^M$ precomputed, initialization uses $O(M + D + Q^2)$ rational operations after construction of the weights. Processing uses

$$O(QD + n_c Q)$$

operations and comparisons, instead of $\Theta(n_c Q^2)$ for dense cached scores. Since $D \leq \min(M, n_c Q)$, the dominant score saving can approach a factor Q in sparse support. Dense support retains the same arithmetic order; no universal finite speedup is implied.

Streaming the future products by exact division avoids the Q by n_c suffix table. The shared power table, loads and sparse factors occupy $O(M + Q^2 + D + Q)$ rational entries. The delivered adjacency-matrix prototype also stores and scans its supplied matrix, so its $O(n_r n_c + n_r Q)$ input/preprocessing cost, including greedy row-label scans, is not hidden. A sparse edge-list implementation could avoid that matrix but is not claimed here.

For the inherited choice $z = (hC + 1)/(hC)$, products have polynomial bit length: the total edge-weight degree is at most M, and at most n_c factors contribute denominators Q. Exact additions, reduced fractions, divisions and comparisons consequently remain polynomial-time. These arithmetic counts do not claim the same ratio of bit times.

6.3 Measured component results

All three score paths were implemented afresh: literal sums, dense cached sums, and sparse increments. Preprocessing and power tables are shared, and the dense baseline also receives streaming future products, strengthening the comparison. Imported executable code was not run. Normal and optimized Python each pass 140 fixture/label configurations with identical scores, choices, loads and final potentials; 470 independently enumerated completions verify the conditional means, and four invalid inputs are rejected.

Q	Columns	Rows	Column degree	Cached ms	Incremental ms
8	32	16	2	6.983	1.925
16	48	32	2	37.156	3.908
24	64	48	2	87.676	7.040
24	64	48	8	151.508	32.844
12	32	24	24	36.244	24.179

The recorded median ratios are 3.63, 9.51, 12.45, 4.61, and 1.50. Candidate-term counts change respectively from $2048, 12288, 36864, 36864, 4608$ to $504, 1488, 3048, 10584, 3924$. The benchmark uses three repetitions in alternating order, includes setup and invariant checks, and omits verbose trace generation equally. The recorded environment is Python 3.12.14 on Linux x86-64; exact samples, deterministic fixture generation and counts are retained. These selected sparse-to-denser fixtures are not representative SAT workloads. Timing variation is expected, and the measurements do not establish end-to-end SAT speedups or a practical asymptotic crossover.

7 Centered antipodal packet extraction

The following improvement is independent of the packing, taper and Cauchy-mode certificates. It replaces only the packet-entry coefficient extraction, not the mode sum or the SAT solver. The latest inspected reports retain the earlier streamed forward-difference routine. We provide a same-interface alternative, including the necessary characteristic-two precision accounting, an independent implementation, and comparisons against a forward-grid implementation already optimized by Gray traversal. The underlying polarization, Glynn, and root-filter ideas are classical; no priority is claimed for these identities. For context, see Glynn's *The permanent of a square matrix* (2010) and Aaronson–Hance's generalized Glynn treatment (2012).

7.1 Generic half-grid extraction with correct guard precision

Homogenize a packet product of R affine forms by adding a constant coordinate of target exponent $R - |\beta|$; return zero if $|\beta| > R$. After dropping zero-exponent coordinates, write

$$H(Z) = \prod_{r=1}^{R} \sum_{j=1}^{t} a_{rj} Z_j, \qquad \sum_{j=1}^{t} \beta_j = R, \qquad G = \prod_{j} (\beta_j + 1), \quad B = \prod_{j} \beta_j !.$$

The empty product $R = 0$ is handled directly as the constant one. All β_j are now positive. The coefficient ring is the source's finite free quotient

$$A_p = (\mathbb{Z}/2^p\mathbb{Z})[U, V]/(f(U), V^\ell),$$

with f monic; the argument allows nilpotents and does not require A_p to be a field.

Proposition 7.1 (Paired centered differences)**.** *For $R > 0$, let $\mathcal{H}$ contain one representative of each pair $\{\gamma, \beta - \gamma\}$ in $0 \le \gamma \le \beta$, omitting the central point when every β_j is even. Then*

$$2^{R-1} B[Z^\beta]H = \sum_{\gamma \in \mathcal{H}} (-1)^{R-|\gamma|} \prod_j \binom{\beta_j}{\gamma_j} H(2\gamma - \beta). \tag{P1}$$

There are exactly $\lfloor G/2 \rfloor$ evaluations. If $v = v_2(B)$, the coefficient modulo 2^p is recovered exactly by evaluating the right side at precision $p + R - 1 + v$, dividing canonical integer coordinates by 2^{R-1+v}, and multiplying by $(B/2^v)^{-1} \bmod 2^p$.

Proof. Apply the difference of step two and order β_j in each variable, starting at $-\beta_j$. A monomial of total degree R vanishes if any exponent is below β_j; the only survivor is Z^β, with multiplier $2^R B$. Homogeneity gives $H(-(2\gamma - \beta)) = (-1)^R H(2\gamma - \beta)$, while the binomial sign changes by the same parity. Paired summands are therefore equal. The unpaired central value is $H(0) = 0$. Dividing the integral identity by two proves (P1). Its validity in the free integral quotient proves the stated coordinatewise divisibility. Keeping the additional $R - 1 + v$ bits, rather than illegally dividing a residue at precision p, recovers the answer. □

The representative set is a disjoint union of at most t boxes: the first noncentral coordinate is below its midpoint, earlier coordinates are central, and later coordinates are unrestricted. Reflected mixed-radix Gray traversal changes one coordinate by one at a time. Keeping the R current row sums updates them in $O(R)$ ring operations per evaluation, instead of recomputing t terms per row. No grid-indexed table is retained. The input and preprocessing are polynomial-sized, and auxiliary streaming state is polynomial in $R, t, p, \deg f, \ell$. The extra precision is at most $2R - 1$ bits. Both this routine and the strong forward-Gray baseline use $O(RG)$ ring operations up to polynomial setup; the half-grid saves evaluations, not a new exponential order. The additional bit width means a wall-clock improvement is not a universal consequence of the evaluation count.

7.2 A cube-root profile using only unit division

For a better special profile, suppose $\omega^2 + \omega + 1 = 0$ in A_p. For positive target exponents $\beta_j \in \{1, 2\}$, choose an index k of minimum exponent and put $Z_k = 1$. Then

$$[Z^\beta]H = 3^{-(t-1)} \sum_{\gamma \in \{0,1,2\}^{t-1}} \omega^{-\sum_{j \ne k} \beta_j \gamma_j} H(Z_k = 1,\ Z_j = \omega^{\gamma_j}\ (j \ne k)). \tag{P2}$$

All arithmetic is at precision p: 3 is a unit, so no factorial or power-of-two division occurs. To prove (P2), apply character orthogonality to a monomial Z^α. A survivor has $\alpha_j = \beta_j + 3m_j$ with $m_j \ge 0$ for $j \ne k$. Homogeneity forces $\alpha_k = \beta_k - 3\sum_{j \ne k} m_j$. Since $\beta_k \le 2$, nonnegativity implies every $m_j = 0$. Thus precisely the requested coefficient survives.

For $\beta = (2, \ldots, 2)$, (P2) takes $3^{t-1} = G/3$ evaluations, versus $(G - 1)/2$ centered evaluations and G forward evaluations, with *zero* extra precision bits. For many exponents one, the ternary grid may instead be larger; the implementation dispatches by the proven counts and never assumes this special-profile benefit is universal. A single surviving coordinate is handled directly by multiplying its row coefficients.

This does not require enlarging the source's coefficient ring. Its residue field has degree $e = 4\lceil \log_2 s \rceil$, so it contains $\mathbb{F}_4$. A nontrivial root of $X^2 + X + 1$ in that subfield lifts uniquely to every precision, since its derivative $2X + 1$ is a unit. If no root is supplied, exhaustive residue search uses at most $2^e < 16s^4$ candidates, hence polynomial time in s; this is a conservative construction bound, not a

claimed practical search. Subsequent Hensel lifting is polynomial in the ring dimension and precision. The supplied implementation accepts and verifies a root, and supplies a bit-by-bit Hensel lift from a residue root. Benchmarks state explicitly whether that lift is timed; they do not hide a residue-field search.

Creation precision. For both methods the row coefficients are first canonically represented at the tensor's immutable creation precision p. Guard bits in (P1) are internal computation only. Return canonical coordinates modulo 2^p, multiply the prescribed source-mode powers as before, and retain that creation-time representative in later recursive calls. The root filter works in the existing ring and changes no recursive tensor identity.

7.3 Verification and honest same-operation measurements

The independent code executes no imported report program. A direct coefficient multiplication oracle checked 6,737 cases against three kernels, giving 20,211 exact equalities. Cases include exhaustive small binary matrices, noncanonical integer inputs, free monic quotients, nilpotents, and mode-weighted synthetic factors. Additional checks cover 180 cube-root cases, 32 degree-four Hensel fixtures, 212 adaptive-dispatch cases, 710 antipodal partitions and 1,014 Gray boxes. The same suite passes normally and under `python -O`. Negative controls detect insufficient guard precision, an invalid character root and changing creation precision. The exponential-table direct oracle is a small-case verifier, not the streaming algorithm.

Measurements use Python 3.12.14 on an AMD EPYC 9V74 shared Linux executor, one untimed warmup and the median of three timed calls, with alternating method order. Generation of synthetic inputs is excluded; canonicalization, preprocessing, extraction and final division are included. Timings are not isolated-CPU measurements and have no asserted statistical significance. The baseline already performs Gray row-sum updates; it is not a claim of a best possible permanent implementation. Its zero-origin evaluation could trivially be skipped, saving one out of G points.

Profile	p	Forward Gray (s)	Half grid (s)	Ratio
12 exponents one	64	0.05786	0.02698	2.14
16 exponents one	64	1.06319	0.51464	2.07
18 exponents one	256	5.41105	2.71399	1.99
16 exponents one	1024	2.56917	1.11908	2.30
Eight exponents two	64	0.09234	0.05321	1.74
Six exponents three	256	0.06861	0.03437	2.00
$(1, 1, 2, 3, 4, 5)$	256	0.02239	0.01435	1.56
10 ones, quotient dimension 4	64	0.06762	0.03738	1.81
Six twos, quotient dimension 4	64	0.05363	0.02808	1.91

The cube-root table compares all methods in *the same ring*, never a scalar baseline against an extension-ring candidate. Every profile is $(2, \ldots, 2)$. Here $e = \deg f$ and the full dimension is $e\ell$.

(R, e, ℓ, p)	Forward (s)	Half (s)	Cube (s)	Cube + lift (s)
$(12, 2, 1, 64)$	0.03206	0.01833	0.01100	0.01178
$(16, 2, 2, 64)$	0.68721	0.36056	0.24594	0.23389
$(14, 4, 1, 64)$	0.21847	0.10843	0.07282	0.06280
$(12, 4, 2, 256)$	0.15321	0.08484	0.05662	0.05916

The cached-root cube routine is 2.71–3.00 times faster than forward Gray and 1.47–1.67 times faster than the centered routine in these four fixtures. The separate "Cube + lift" timings charge fresh Hensel

lifting on every call; residue roots are supplied in both cases. The fact that some lift-inclusive medians are smaller illustrates executor/timing noise, not negative overhead. A deliberately adverse eight-ones profile in the degree-two ring used 2,187 cubic evaluations versus 128 centered evaluations: the cubic routine took 0.07935 s versus 0.00416 s, about 19.1 times slower. The adaptive routine correctly selected the half-grid. This negative case is why (P2) is selective, not a universal replacement.

Scope of the result. These are measured improvements to a genuine, exact packet-entry operation, with independently checked formulas and no increase to the recursion's space class. They are not full-SAT timings, do not show the frequency of favorable profiles in real SAT contractions, and do not improve the $n \log\log n/\sqrt{\log n}$ global logarithmic order. In the current asymptotic schedule packet-entry work is already lower order. A production integration should benchmark the dispatch on actual generated packets rather than extrapolate these synthetic speed ratios to a solver.

8 What remains open

None of these results supplies the structured summation or compact factorization required by the earlier stronger-order targets. The precision proposal's $n \log\log n/\log n$ exponent requires a recursively valid summation algorithm for genuine high-arity group tensors. Small rank alphabets alone do not provide bounded-degree factors. The scalar parity proposal's $n(\log\log n)^{3/2}/\log n$ target requires its unsupplied compact-factorization interface, including exact invalid-index zeros and immutable creation precision. The prior degree-kernel route requires both a reduced global deformation degree and a mode-family count measured in the compressed wire universe, closed under every recursive child.

These obligations cannot be discharged by lowering a leaf threshold, quoting small-instance timing results, or splitting an arbitrary high-arity tensor into equality nodes. Nor is a non-polynomial cache a polynomial-space result. The alternate manuscript's graded crossing construction is a distinct route with the same fixed-width logarithmic order; its claimed unpublished formalization is not independently certified here. The present coefficient improvements are genuine upper-bound improvements on the stated normalized interface, not lower bounds on other possible schedules or algorithms.

9 Evidence and reproducibility

The accompanying archive contains this editable TeX, the rendered PDF, selected immutable source reports and provenance ledgers, the authored exact-check programs, recorded component measurements, and replay scripts. Imported source files are evidence only: the build compiles only `report/report.tex` and its authored sections. Source ZIPs, build caches and page-render images are excluded.

The recorded verification covers the following finite obligations:

- Exact rational capacity and coefficient checks for the 64 and 24 polynomial-space schedules, including deliberate false-capacity and false-threshold controls that must fail under both normal and optimized Python.
- Exact rational 38/34 and 14.3/12.4 cached certificates, their block counts and geometric recurrence; false smaller thresholds are rejected in both modes.
- The 200-configuration Cauchy-feature audit described in Section 3.
- The balancer's 140 configurations and 470 enumerated completions, plus an independently written audit with 696 fixtures, 3,480 candidate means and 7,600 completion evaluations. The latter includes all 2 by 2 matrices over $\{0, 1, 2\}$ for $Q = 1, 2, 3, 4$ and two rational-base scales.
- The packet-extraction checks described in Section 7, including independent direct-product coefficients, finite free quotient rings, precision guard failures and creation-precision controls. A separately written

identity audit adds 183 centered cases, 63 cube-root ring cases and 1,572 monomial orthogonality checks; it does not import either candidate or upstream executable code.

Every failure predicate in the authored checkers uses an explicit exception rather than a removable Python assertion. Normal and `-O` outputs agree after excluding elapsed-time fields. Recorded benchmark samples are retained; replay validates exact outputs but does not require identical wall-clock timings.

The bundle manifest covers the complete payload inventory with SHA-256 and byte counts. `verify_bundle.py` rejects unsafe names, duplicate members, symlinks, excess expansion, inventory changes and altered payloads. It also checks selected-source hashes against the retained acquisition records; the acquisition ledgers, not a freshly downloaded archive during replay, establish source provenance. Negative controls exercise rejection paths under normal and optimized Python. `replay_archive.py` unpacks to a fresh directory, runs only the authored finite checks in both modes, rebuilds the PDF without shell escape or network access, and compares extracted text and every rendered page's pixels with the sealed PDF. Finite checks are regression evidence, not a formal proof of the asymptotic results. Rebuilding on a different TeX/font distribution may change pixels without changing the mathematics.

References

[1] OpenAI, *The Exponential Time Hypothesis is False*, manuscript source, vendored snapshot. Immutable main source.

[2] OpenAI, quantitative contraction appendix. Immutable appendix.

[3] OpenAI, quantitative reduction appendix. Immutable appendix.

[4] ProveIt, canonical SubexponentialSAT project and external-research index. Immutable canonical project.

[5] ProveIt, eight new proposal archives under `docs/incoming`. Immutable incoming directory. Exact archive names, Git blobs, SHA-256 values and selected members are in the evidence ledger.

[6] *Full Crossing-Block Cauchy Modes*, incoming `eth_joint_cauchy_2026-10-26.zip`, Sections 3–6 and cached-block construction. Closing canonical report. Retained as `evidence/new/joint/report.tex`.

[7] *Asymmetric packet balancing for subexponential SAT*, incoming `eth_asymmetric_packet_balancing_2026-10-26.zip`. Closing canonical report. Retained report and annulus-mode appendix under `evidence/new/asymmetric/`.

[8] *Packet refinement*, incoming `eth_packet_refinement_2026-10-26.zip`; exact cached row-sum implementation inspected as data. Retained report under `evidence/new/packet/`.

[9] Earlier seven canonical research packets. Selected full reports and manuscript sources are retained under `evidence/reports/` and `evidence/openai/`; exact paths and provenance are in `evidence/selected_sources.json`.

[10] *Review and refinement of the SubexponentialSAT proposals*, second-round report, incoming `3sat_round2_refinement_20261027.zip`. Retained main text and derivation sections under `evidence/new/round2/`.